

COURSE NUMBER: CP1212

COURSE TITLE: Introduction to JavaScript

COURSE DESCRIPTION:

JavaScript is among the most powerful and flexible programming languages available and one of the most popular languages used in web development. This course will introduce students to the programming fundamentals of the JavaScript language, where students will learn basic programming constructs to add various functionality to webpages. Then the student will explore jQuery one of the most popular libraries used to help in JavaScript development.

PREREQUISITES: None

CO-REQUISITES: None

CREDIT VALUE: Three (3)

COURSE HOURS PER WEEK: Two (2)

LAB HOURS PER WEEK: Two (2)

REQUIRED TEXT:

Delamater, M (2024) *Murach's Modern Javascript*. Murach Publishing Frenso, CA Print
ISBN: 9781943873142, E-Text ISBN: 9781943873159

LEARNING RESOURCES:

[JavaScript Tutorial](#)
<http://www.tutorialspoint.com/javascript/>
[jQuery website](#)
[jQuery Plugin Registry](#)
[jQuery UI website](#)

MAJOR TOPICS:

- 1.0 Introduction to JavaScript
- 2.0 JavaScript Basics
- 3.0 The DOM, Images and Timers
- 4.0 Working with Legacy JQuery
- 5.0 Using Legacy JQuery plugins

LEARNING OBJECTIVES:

1.0 Introduction to JavaScript

- 1.1 Describe the process that a browser uses to load a web page that includes JavaScript
- 1.2 Use a browser to run JavaScript applications that are on the Internet, on your local computer or a local server
- 1.3 Use the browsers developer tools to find JavaScript statements that can cause an error in a JavaScript application
- 1.4 Use a text editor or IDE to edit JavaScript files
- 1.5 Modify a HTML file by adding JavaScript
- 1.6 Create JavaScript Identifiers
- 1.7 Modify JavaScript code by adding comments
- 1.8 Use JavaScript primitive data types
- 1.9 Create JavaScript code that declares constants and variables
- 1.10 Create JavaScript code that utilize arithmetic operators
- 1.11 Create JavaScript code that utilizes built in JavaScript methods

2.0 JavaScript Basics

- 2.1 Describe conditional expressions
- 2.2 Use logical operators to evaluate compound conditional expressions
- 2.3 Use conditional statements to control the flow of a program
- 2.4 Use looping structures to control the flow of a program
- 2.5 Use a JavaScript array and its properties within a program
- 2.6 Create Number and String objects in a program
- 2.7 Create the Date object using JavaScript
- 2.8 Create functions in JavaScript
- 2.9 Use function expressions and arrow functions
- 2.10 Work with event handlers in JavaScript
- 2.11 Rewrite code to fix syntax, runtime and logic errors
- 2.12 Create a simple test plan
- 2.13 Use the `console.log()` method to trace the execution of an application

3.0 The Dom Images and Timers

- 3.1 Develop a JavaScript application that manipulates the DOM and:
 - 3.1.1 Works with forms
 - 3.1.2 Works with controls
 - 3.1.3 Adds nodes to the DOM
- 3.2 Use the browsers developer tools to view the changes that JavaScript had made to the DOM
- 3.3 Preload images for an application where the images aren't automatically loaded by the web page

- 3.4 Use one-time timers in a JavaScript application
- 3.5 Use interval timers in a JavaScript application

4.0 Working with Legacy JQuery

- 4.1 Add the JQuery library to a web page so that it can be used in a JavaScript application
- 4.2 Use JQuery to develop common DOM scripting applications including:
 - 4.2.1 Email Lists
 - 4.2.2 Faqs
 - 4.2.3 Image Swaps
 - 4.2.4 Image Rollovers
- 4.3 Use JQuery to add effects and animations on a web page
- 4.4 Use JQuery selectors, methods, and events methods to work with forms and controls in a JavaScript application

5.0 Using Legacy JQuery Plugins

- 5.1 Use JQuery plugins and UI widgets in a JavaScript application
- 5.2 Describe a JQuery plugin
- 5.3 Use a variety of JQuery plugins including:
 - 5.3.1 Lightbox
 - 5.3.2 bxSlider
 - 5.3.3 Cycle 2
- 5.4 Use a variety of JQuery UI widgets including:
 - 5.4.1 Accoridan
 - 5.4.2 Tabs
 - 5.4.3 Button
 - 5.4.4 Dialog
 - 5.4.5 DatePicker

EVALUATION:

Assignments:	30%
Test(s):	20%
Project:	20%
Final Examination:	30%

DATE DEVELOPED: September 2020

DATE REVIEWED:

REVISION NUMBER: 1

DATE REVISED: June 2025

Effective PeopleSoft Date: September 2025

Note to instructor: Check PIRS to ensure this outline is the most current version.